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ORIGINAL RESEARCH PAPER

Acoustic fingerprint in vehicle manufacturing as a basis for future applications

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ABSTRACT

The spread of new energy vehicles has opened a new chapter in premium-class acoustics. In e-mode, hitherto inaudible noises – suppressed by traditional engines – have become audible. Therefore, it is necessary to make subjective noises objective. The first step is to make them comparable by measuring. The result of the sum of these spectrograms is an acoustic fingerprint of the vehicle.

The next step is to add them to a database. Early failure prediction and the avoidance of redundant service activities by using automated event detection led to financial savings. Data processing with machine learning opens space for car workshop support applications, and it is also possible to prevent critical failures that may begin with the help of predictive models (predictive service).

KEYWORDS

whole vehicle acoustic analysis, acoustic fingerprint, prediction by machine learning, acoustic database, predictive repair

1. INTRODUCTION

The lightweight structural materials and increasingly complex technical solutions used today in the automotive industry place an increasing burden on engineers during the design phase, testing, and possible fault analysis or service activity [1]. The result of the spread of plug-in hybrid and battery electric vehicle technologies is that all noises are (more) audible in e-mode than before [2]. At the same time, there is a growing shortage of qualified mechanics. All these require automation and simplification of vehicle acoustic tests [3, 4].

There is a growing demand for the perfect audiovisual experience while traveling and for the widest possible use of communication options too. Of course, this can only be achieved with outstanding vehicle acoustics. Most mechanical failures and, obviously, the operation and movement of the vehicle cause noise even in flawless conditions, so it is obvious to use acoustics as an important design and diagnostic tool during the design and production process of vehicles [5, 6]. See more detail below, where the acoustic-driven construction is also considered [7].

Already during the design phase, engineers need to consider the acoustic properties of components, component groups, and the whole vehicle under quality standards and customer requirements. However, despite modern and high-tech design work, defects may remain in the vehicle as an acoustic system for the reasons mentioned above. This is further complicated by the production dispersion of parts, which can also be reduced, but above a certain level, it is no longer economical.

1.1. Actual used methods for creating acoustic fingerprints

The acoustic methods and devices used in the automotive industry, especially in manufacturing and analysis, must be as cheap and simple as possible. This allows measurements with a large sample number with the lowest possible pre-training. For deeper

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analyses, standardization, comparability of results, and measurement methods, e.g., with product development or methods/results of suppliers or other factories is important. In the following section, details will follow the most used vehicle acoustic analysis methods [8, 9].

Further methods in car manufacturing/service:

- A simple numerical measure such as sound pressure level;
- Frequency analysis like FFT (spectrum) reference;
- Frequency-time analysis;
- Tests related to other vehicle parameters (e.g., temperature, speed);
- Vehicle shaker bench test;
- Dynamic acoustic test (moving vehicle) [10].

1.2. Speed-related tests

This is the so-called order analysis, the purpose of which is to determine and localize the origin of the noises generated by the system by analyzing multiples of given typical frequencies. An example could be the internal combustion engine (crankshaft-camshaft speed ratio is always 2:1 in the case of a four-stroke engine, or harmonic components that can be counted from the contact ratio of specific auxiliary drives, for example, air conditioning compressor, generator, fuel pump, etc.). An even more typical example is the Noise, Vibration and Harshness (NVH) analysis of torque converters in the knowledge of gear ratios of the given gears [11]. So, this important method is only used for root cause analysis purposes but not for acoustic fingerprints.

1.3. Psychoacoustics

Since the human ear does not perceive different frequency noises, only in the 20 – 20,000 Hz range, higher frequencies are more disturbing, the frequency-time-sound pressure spectrogram has been converted to the frequency-time-loudness one. The psychoacoustic models currently used attempt to adopt this algorithm of human sound perception. Figure 1 shows the psycho-acoustical behavior of a vehicle's rear lid. The original frequency-time and acoustic pressure chart has been changed for a frequency-time and loudness one, which models better how a human (vehicle user) reacts to this noise: the most disturbing middle-high frequency area is marked with red in the HEAD Artemis software.

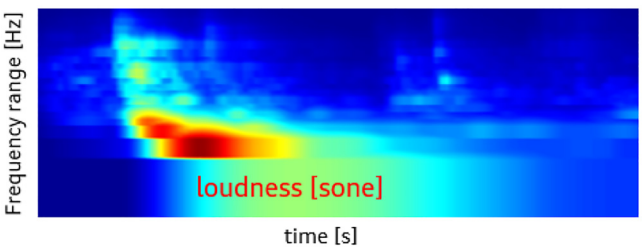

Fig. 1. Principle of psychoacoustic frequency-time spectrogram: the sound pressure in dB has been changed to loudness in sone, and the colored areas show different levels of loudness (*Source*: Authors)

2. METHODOLOGY: CREATING ACOUSTIC FINGERPRINTS IN THE PRAXIS

This test is a typical comfort-acoustic analysis, where noise emitted by various actuators or other moving vehicle elements is usually examined. The method of documentation or display here is a time-frequency plus sound pressure or rather loudness diagram, whose mathematical background is the Fast Fourier Transform (FFT). The measurement series was realized using the SQuadriga II device from HEAD Acoustics with software HEAD Artemis Suite 13.2 software and HEAD Acoustics HMS IV Artificial Head on new midsize SUV cars.

For the description of acoustic fingerprints of vehicles, the following 12 systems have been defined; these are most frequently complained about by internal and external customers: front and rear doors, both sides; electric tailgate; electric window lifts front, left, and right sides; panoramic sliding roof; glove box; both front (electric) seats; electric parking brake.

During the first measurements, the measured values were plotted in HEAD Artemis software in three ways:

1. Sound pressure-time chart;
2. Fast Fourier Transformation (FFT), which is a frequency-time spectrogram with different colors for given sound pressure values;
3. Frequency-time spectrogram with different colors for given loudness values (see Fig. 2).

3. RESULT: ACOUSTIC FINGERPRINT DATABASE FOR FURTHER ACTIVITIES

In the mentioned way, the 12 most important systems from a comfort acoustic point of view mentioned above should be recorded. This would have meant 36 diagrams, so to simplify it; it was decided to keep only one of the three representations. For support of future use with machine learning and considering the subjectivity of human loudness perception, it was decided, to keep only the frequency-time/loudness spectrogram, see examples in Fig. 3.

When testing the closing noise of doors for standardized purposes [12], the question of the defined closing speed arose, since increasing the closing speed naturally results in higher noise levels, see Table 1. It can be seen, that when measuring at an undefined closing speed, the standard deviation of loudness values increases greatly, which would make accurate evaluation impossible in this case. (Based on experiences, an average person can distinguish a difference in sound pressure of about 3 dB; the minimum and maximum values of the measurements were 6 dB).

In summary, the acoustic fingerprinting method based on the 12 vehicle actuator and door/cover systems is suitable for several applications:

- To classify a claimed car against the reference values including tolerances;

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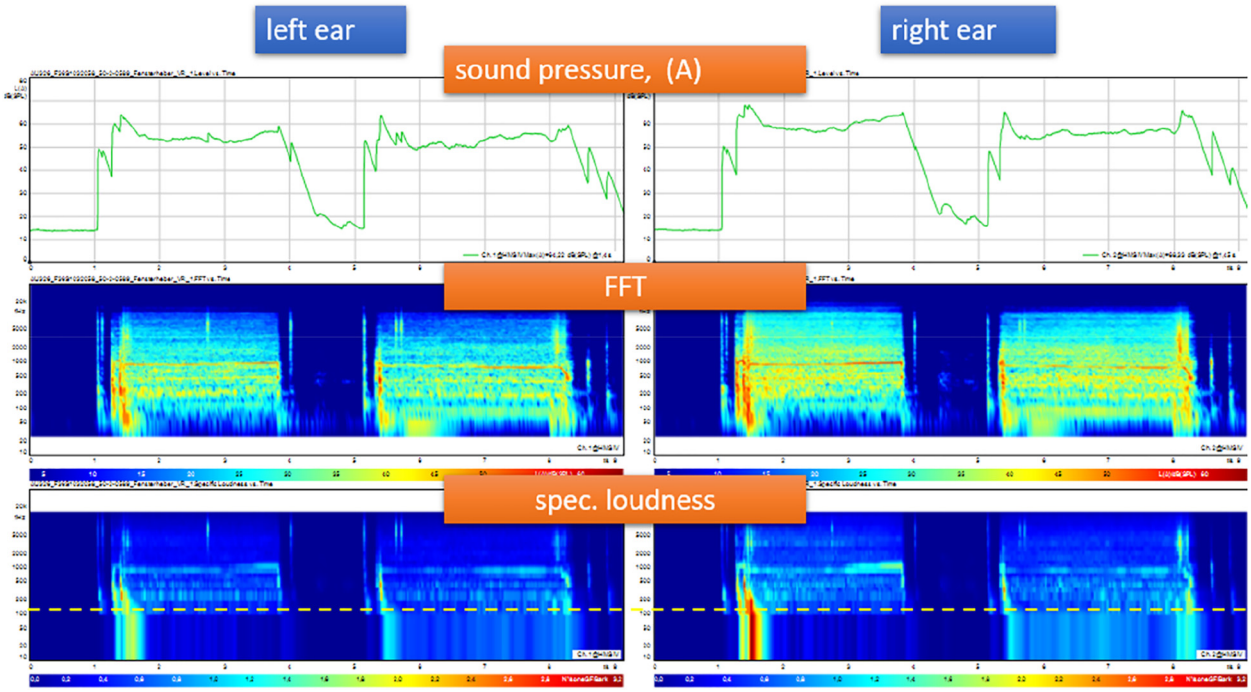

Fig. 2. Results of front window lift measurements left vs. right side (principle, scale is not relevant): sound pressure (Y)/time (X) chart (top), FFT: frequency (Y)/time (X)/sound pressure (colors) spectrogram in the middle, frequency (Y)/time (X)/specific loudness (colors) spectrogram bottom (*Source*: Authors)

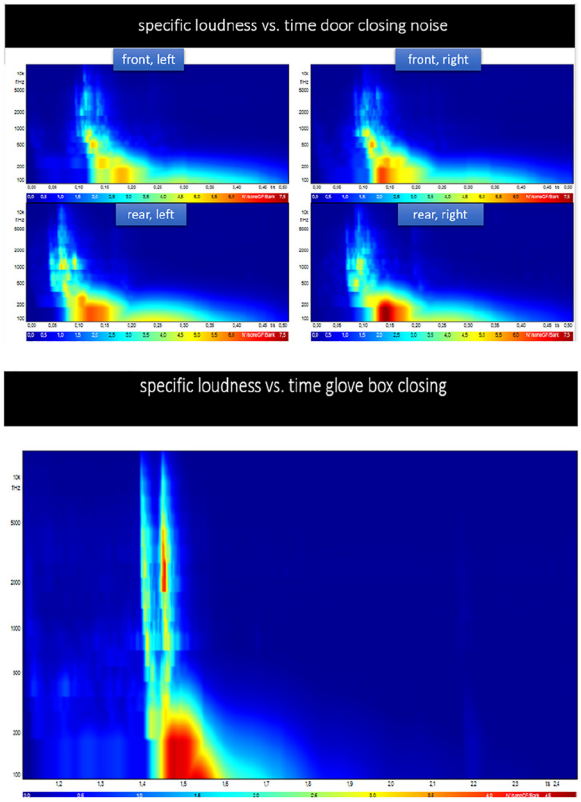

Fig. 3. Example for spectrograms for door and glove box closing noises: (principle, scale is not relevant): Y-axis: frequency (Hz), X-axis: time (s), colored fields: specific loudness (sone) (*Source*: Authors)

Table 1. Summary of measure series of driver door: an unspecified closing speed results 1.6 dB(A), then 1.21–1.23 m s^{-1} closing speeds (specification) show only 0.2 dB(A) standard deviation

Group name	Max dB(A)	Mean value dB(A)	Standard deviation	Speed (m s^{-1})
Tür_VL_1	80.4			1.37
Tür_VL_2	77.8			1.22
Tür_VL_3	80.0			1.32
Tür_VL_4	75.3			1.06
Tür_VL_5	74.5			0.96
Tür_VL_6	77.0			1.19
Tür_VL_7	78.0			1.23
Tür_VL_8	77.7			1.22
Tür_VL_9	77.5			1.21
Tür_VL_10	79.2			1.27
Tür_VL_11	76.2			1.11
Tür_VL_12	78.9			1.27
Tür_VL_13	80.2			1.32
Tür_VL_14	77.5			1.20
Tür_VL_15	78.7			1.25
Tür_VL_16	78.5			1.25
Tür_VL_17	80.5			1.39
Tür_VL_18	77.9			1.22
Tür_VL_19	78.3			1.24
Tür_VL_20	76.46			1.15
Average		78	1.6	
Tür_VL_2	77.8			1.22
Tür_VL_7	78.0			1.23
Tür_VL_8	77.7			1.22
Tür_VL_9	77.5			1.21
Tür_VL_18	77.9			1.22
Average		77.8	0.2	

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- Objective comparison with competitors or old models with their successors;
- Comparing vehicles of different categories manufactured by the same brand for feature planning purposes (e.g., the operating noise of a class B or C car should not be higher than that of a cheaper, class A vehicle).

The sounds emitted by approved prototype vehicles and their various systems are subjectively considered good by the top management: there is a quality level standard for future claim evaluations - this is a method from the past. After analyzing competitors, made objectively through measurements (tolerances are determined and considered), the result is the so-called acoustic fingerprint - that is the present.

The feature matrix plots loudness spectra as a function of different environmental/setting parameters. These include vehicle speed, wind speed, outside temperature, test track section, engine speed, door lock setting depth, sealing rubber hardness, etc., so any parameter with an impact on measured sound pressure or loudness level [13].

4. FUTURE NVH DETECTION AND REDUCTION STRATEGY IN CAR MANUFACTURING

The future-oriented development is to prepare the acoustic fingerprint (spectrogram and curve) for the given components/systems in addition to various influencing factors: this is how feature matrices databases are formed. By comparing the measured spectrograms in the database supported by machine learning with the current measurement result, the given noise can be localized: thus, helping accelerate the analysis and starting activities for predictive repair purposes (see Fig. 4).

The description process of the acoustic behavior of vehicle systems (mostly actuators) is:

1. level: in the acoustic chamber without disturbing noises;
2. level: while driving includes different environmental influences, result feature matrix;
   1. step: measurement series and feature extraction including data classification;
   2. step: feature matrix - acoustic database (see Fig. 5).

As just mentioned, a new approach was introduced in the automotive industry, namely that instead of subjective noise assessment and instrumental measurement limited to complaint analysis, the concept of acoustic fingerprinting was implemented. The focus here is on actuator and door- or window systems that are either close to the driver's ear, emitting noise more disturbing than other noises in the vehicle, or can also generate disturbing sounds when operated by the user. The essence of this method is that the listed systems of cars are subjectively rated as good by management and compared with competitors, and then they are measured in the pre-series. Tolerances are defined for the measured values, which serve as guiding specifications in case of subsequent complaints.

As it has already been mentioned earlier, it is advisable to plan to base the method on classical pattern comparison, like e.g., facial recognition algorithms [14–17]. Here, instead of a saved facial image, the given spectrogram from the database would be compared with the current image. Above a given level of similarity significance; it can be stated that the source of the recorded noise is the same as in the database. For this,

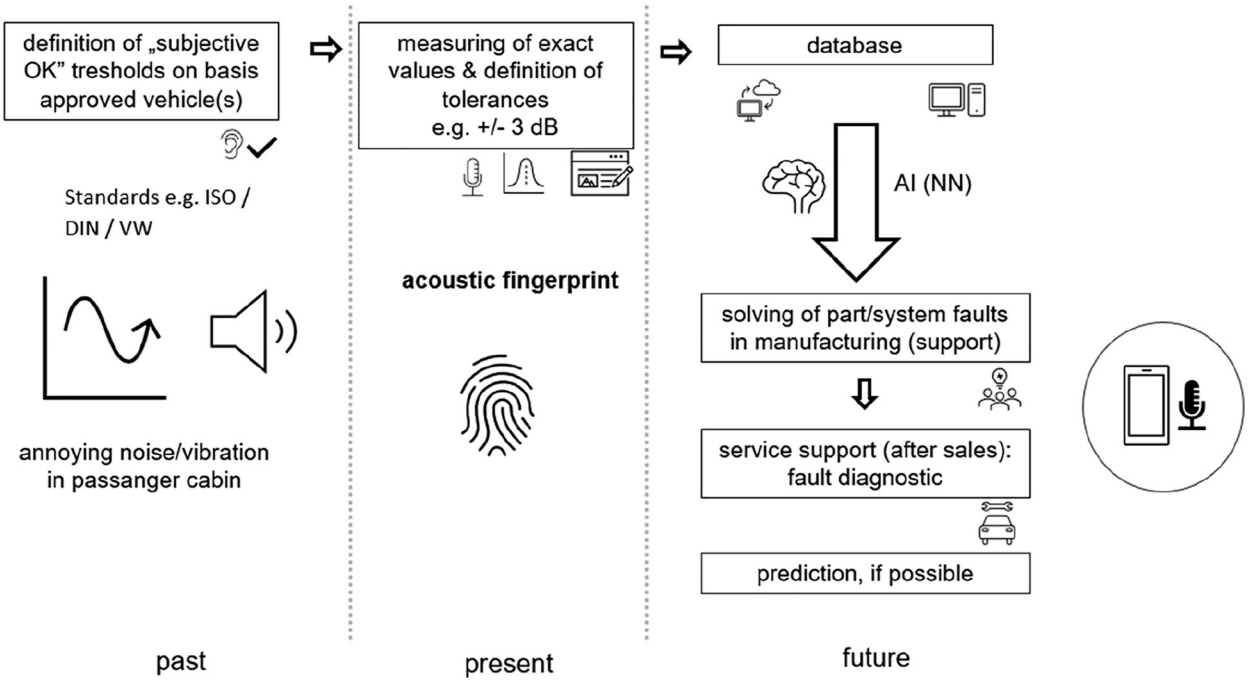

Fig. 4. Past, present, and future activities on NVH analysis strategy (*Source*: Authors)

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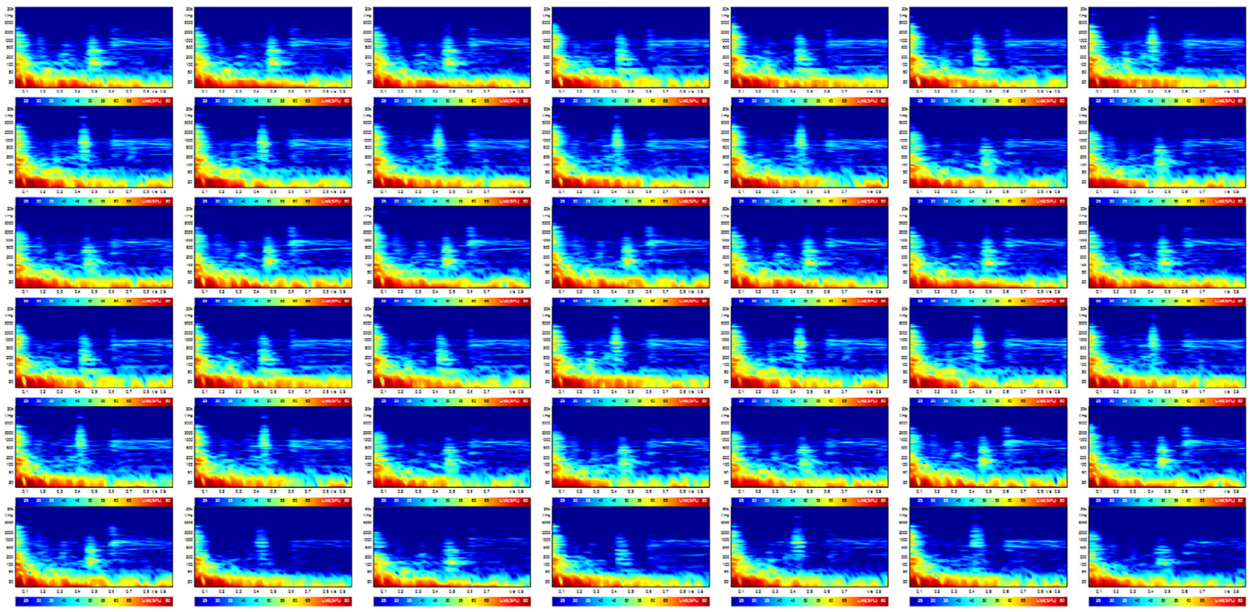

Fig. 5. An example of a 7 × 6 feature matrix: 7 measure series with 6 parameters (*Source*: Authors)

the goal is to integrate this feature into an existing acoustic cell phone app as an extra module. With this method, it can develop the work of analysis technicians to be faster and thus more efficient in case of simple failures in series production. In the car service workshop, it can compensate for any shortcomings in their professional and experience for the staff. The challenge here will be to filter out disturbing background noise for dynamic (driven vehicle) analyses. For the model schema of the machine learning pipeline required for this, see Fig. 6.

It should be also possible to predict and prevent failures in the future based on acoustic records compared with the database in case of noises not auditable yet by humans. The result is then a predictive repair [18].

5. CONCLUSION

Based on the mentioned details, it can be stated that a database containing a sufficient number of acoustic loudness spectrograms can serve as a basis for further developments. For example, in automatic noise assessment, with the help of a mobile phone app, the machine learning model can

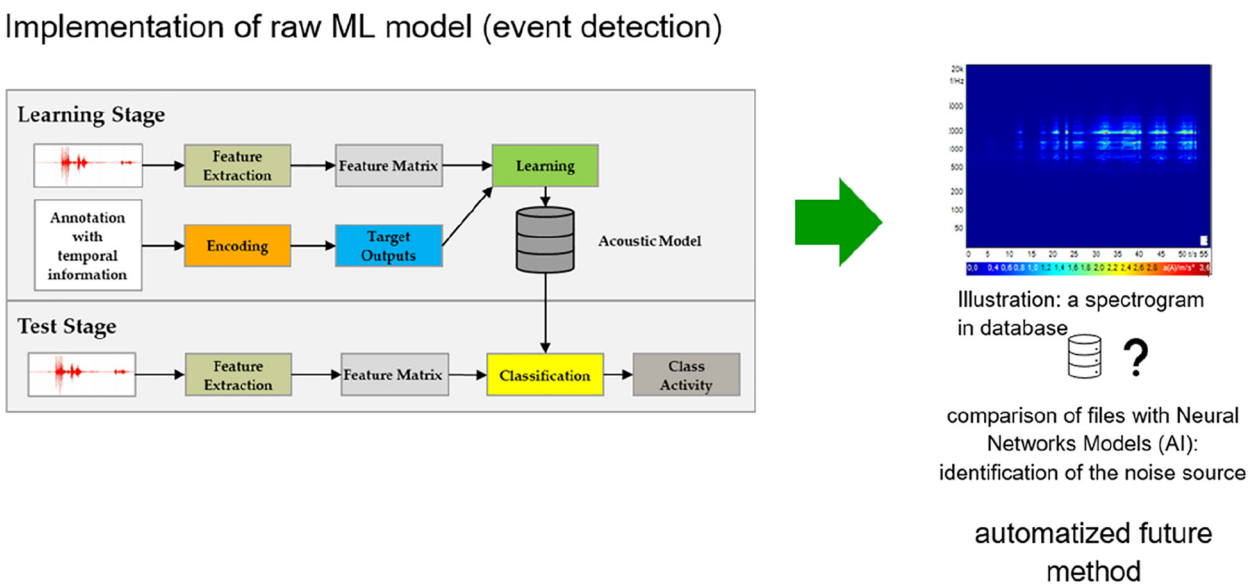

Fig. 6. Based on the raw Machine Learning (ML) model the next step is creating an acoustic database (feature matrix) for neural network models to identify noises (new records vs. data from the database) (*Source*: on the basis of [19])

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compare the recorded sample with the database and advise the investigator on the noise source.

Another application could be a predictive repair, also based on the database, to avoid potential component failures by localizing known noises that are not yet audible to the human ear.

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